

Module 04 (b)

Text Editing with vi

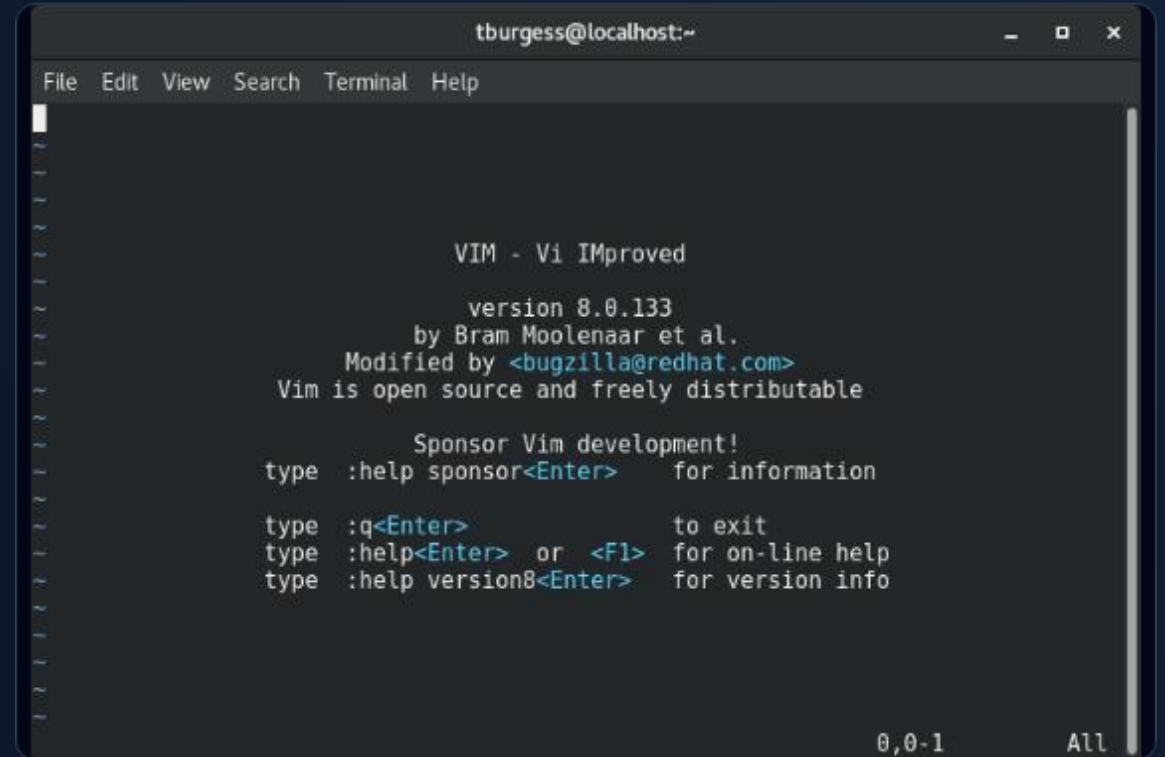


Core Architecture & Modes

Understanding why vi is modal and how modal editing drastically speeds up software development and system administration.

Why Learn vi in Engineering?

- > **Universal Availability:** Pre-installed on virtually every Linux, Unix, and macOS terminal environment worldwide.
- > **Remote Administration:** Essential when editing configuration files on head-less remote servers over SSH.
- > **Lightweight & Fast:** Zero GUI overhead; opens massive log files and codebase files instantaneously.
- > **Keyboard Efficiency:** Designed to keep your hands on the home row without touching a mouse.



A screenshot of a terminal window titled 'tburgess@localhost:~'. The window shows the Vim help screen for version 8.0.133. The text is as follows:

```
VIM - Vi IMproved

        version 8.0.133
        by Bram Moolenaar et al.
    Modified by <bugzilla@redhat.com>
Vim is open source and freely distributable


Sponsor Vim development!
type  :help sponsor<Enter>    for information

type  :q<Enter>                to exit
type  :help<Enter> or <F1>    for on-line help
type  :help version8<Enter>  for version info
```

At the bottom right of the terminal, the text '0,0-1 All' is visible.

The Three Essential Modes



Normal Mode

Default mode upon opening. Keystrokes act as navigation and editing commands (e.g., delete, copy, jump) rather than inserting text.



Insert Mode

Allows typing raw text into the document.

Enter by typing `o`, `O`, or `Ctrl-O`. Leave using

`i`

`a`

`o`

`Esc`



Command-Line Mode

Activated by typing in Normal mode.

Executes file operations, saving, quitting, search/replace, and settings.

Mode Switching & Navigation

Home Row Cursor Controls

Use (**h** (down) **j**), and (right **k** Normal mode **l**) navigate without arrow keys.

Jump quickly using (next **w** ord), (previous **b**), (start of line), and (end **0** of line).



Opening, Saving, and Exiting

Saving & Standard Exit

Return to Normal Mode (), then **Esc** :

`:w` — Write/Save file changes

`:q` — Quit (fails if unsaved changes exist)

`:wq` or `:x` — Save and quit in one step

Force Quit & File Management

Essential emergency and quick commands:

`:q!` — Force quit discarding all changes

`:w filename.txt` — Save as a new file

`vi filename.c` — Open or create file from shell

Text Editing & Undo Commands

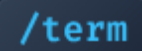
 Insertion Modes: Press to insert **i** before cursor, to append after **a** and to open a new line below **o**

 Character & Line Deletion: Press to delete **x** character under cursor; deletes current line **dd** deletes word. **dw**

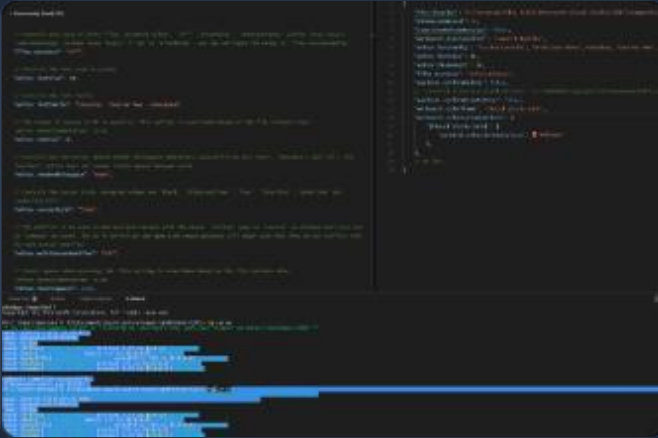
 Undo & Redo Actions: Press in Normal **u** mode to undo previous action; press to redo. **Ctrl + r**

 Replace Character: Press followed by **r** by a letter to instantly replace single character without staying in insert mode.

Copy, Paste & Search Tools

Operation	Command Key	Action Description
Yank (Copy) Line	yy 	Copies the current line into memory buffer
Yank Word	yw 	Copies word from cursor to word ending
Put (Paste) Text	p  	Pastes buffer after (p) or before (P) cursor position
Forward Search	/t 	Searches forward for term ; press n for next,  previous 
Global Substitution	:% 	Replaces all occurrences of old with new in entire file

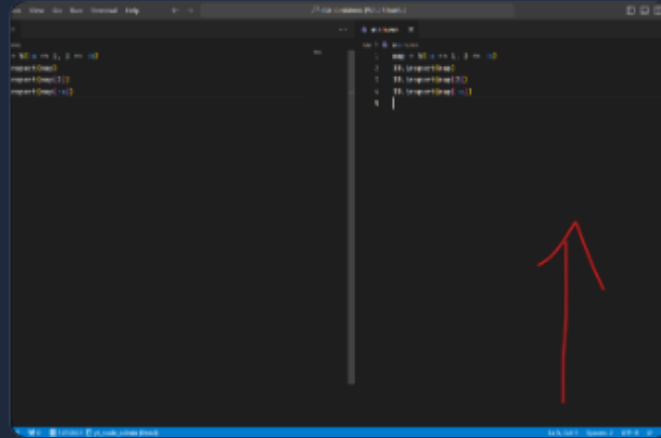
</> Visual Mode & Multi-Buffers



Visual Selection (v)

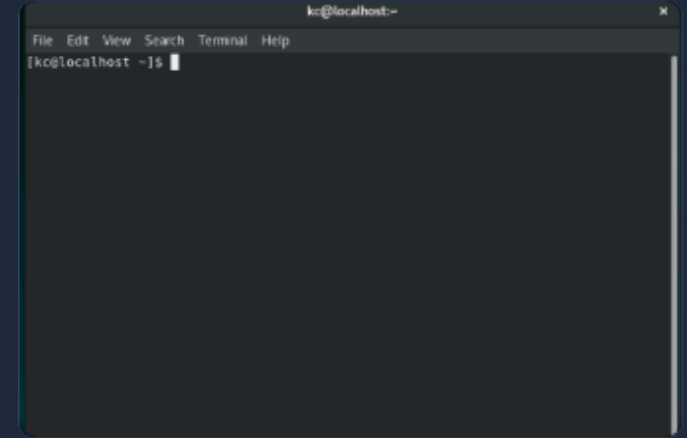
Highlight character blocks with `,w`, whole lines with `,l`, or rectangular blocks with `,r`.

`Ctrl+v`



Window Splits (:sp)

Split screen horizontally with `:sp` or vertically with `:vsp` to edit multiple files simultaneously.



Shell Commands (:!)

Run Linux shell commands directly from inside vi using `!command` without leaving the edit session.

⚡ CLI Productivity Metrics

100%

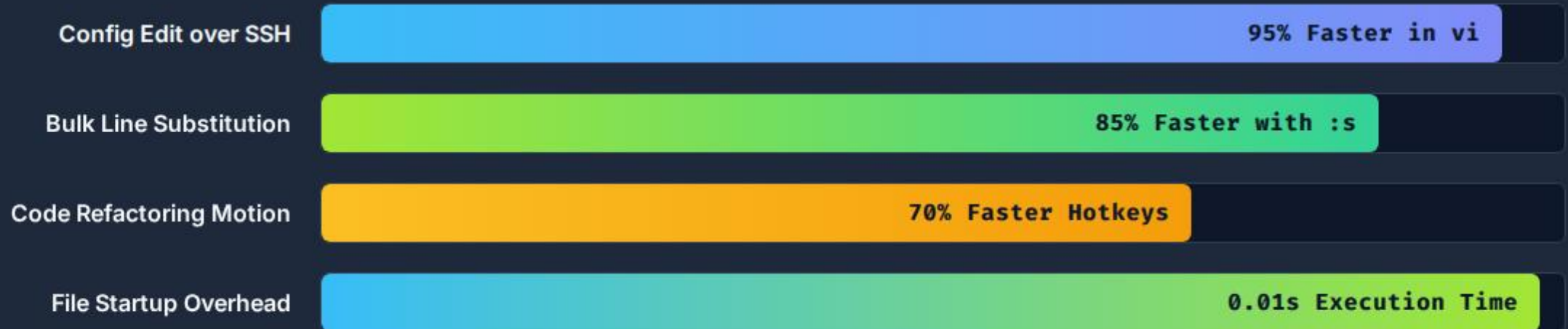
Keyboard-Driven Speed

Eliminating Mouse Latency

Studies show moving a hand from keyboard to mouse takes approximately 0.5 to 1.5 seconds. Over thousands of line edits per day, keyboard-centric modal editing in **vi/vim** saves hours of accumulated context-switching time.

Engineers using modal editing retain uninterrupted mental flow during complex software debugging.

|| Editing Workflow Speed Gain



Comparison highlights the dramatic speed advantages of vi/vim over GUI-based editors when working in Linux command-line environments.

Questions & Practice

Next Steps: Run vimtutor in your terminal to complete today's lab assignment!

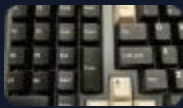
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Image Sources



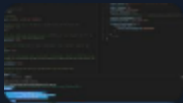
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Source: medium.com



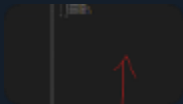
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Source: www.keeptalk.com



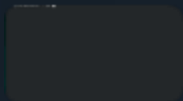
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Source: stackoverflow.com



<https://preview.redd.it/code-and-terminal-window-side-by-side-possible-v0-razqbmd6lpod1.png?width=1910&format=png&auto=webp&s=9270231c7cb9ff534646fb4d9fa7a7d4d57f3da9>

Source: www.reddit.com



<https://www.redhat.com/rhdc/managed-files/sysadmin/2020-02/CPFigure1.png>

Source: www.redhat.com